

Ziyi Sun

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EDUCATION

- **Ocean University of China (985 / Double First-Class)** Sep. 2023 – Jun. 2027 (Expected)
B.Eng. in Mechanical Design, Manufacturing and Automation Qingdao, China
 - GPA: **3.7/4.0** Rank: **1/56** Average: 89.2
 - Selected courses: Advanced Mathematics (97), Linear Algebra (93.5), Probability & Mathematical Statistics (93), Complex Variables & Integral Transforms (94.5), Numerical Methods (96), Mechanical Vibrations (95), Modern Mechanical Design Theory and Methods (100), Underwater Robotics (97.2)

RESEARCH EXPERIENCE

- **Roboparty Lab** Jul. 2026 – Oct. 2026
Frontier Research and Open Source Intern (Mentor: Jagger) Beijing, China
 - Contribute to frontier research and open-source development for embodied intelligence and robot learning systems.

PROJECTS

- **Variable-Length Multimodal Biomimetic Robotic Fish for Deep-Sea Inspection** Dec. 2025 – Dec. 2026
Role: Project Leader | Tools: SolidWorks, multimodal vision & sensing OUC SRDP
 - Designed a biomimetic robotic fish with continuously adjustable body length (1.2–1.5 m) for deep-sea cage cruise and narrow-space inspection.
 - Fused school-fish visual tracking with water-quality sensing for joint monitoring and early warning of aquaculture risks.
- **Cross-Domain Wire-Driven Biomimetic Eel Robot** Dec. 2024 – Nov. 2025
Role: Project Leader | Tools: vision tracking, Gaussian Splatting, damage detection Provincial Innovation Program
 - Developed an eel-like robot with cruising and active depth control for floating offshore wind dynamic-cable inspection.
 - Integrated visual tracking, 3D reconstruction, and damage detection for cable modeling, defect localization, and risk warning.
- **Streaming Evaluation Framework for Deep-Sea Mining** 2025 – 2026
Role: Lead Contributor | Tools: CAD modeling, multi-AUV simulation & benchmarking Research Project
 - Built a unified simulation and benchmarking pipeline comparing centralized and distributed multi-AUV nodule collection strategies.
 - Integrated CAD-based robot models and reproducible evaluation protocols for scalable underwater robot learning.
- **High-Stability Duck-Type Wave Energy Converter with Pendulum PTO** Dec. 2024 – Nov. 2025
Role: Core Member | Tools: AQWA, genetic algorithm, SolidWorks National Innovation Program
 - Improved capture efficiency and output stability via hydrodynamics modeling, shape optimization, and a combined PTO with nonlinear adaptive regulation.

PUBLICATIONS & PATENTS

C=CONFERENCE, S=IN SUBMISSION, P=PATENT, SW=SOFTWARE

- [C.1] **A Two-Stage Learning Framework for Autonomous Obstacle Avoidance of an Eel-Like Robot.**
Ziyi Sun, Luwen Li, Changhong He, Shuo Jiang, Zhuoyan Wang, Haoyuan Cheng*.
IEEE International Conference on Real-time Computing and Robotics (IEEE RCAR), 2026. **Best Paper Finalist.**
Proposed a two-stage framework combining multimodal perception, expert demonstration, PPO, AHTA-CPG, and domain randomization; achieved 70%–90% success over 60 real water-tank trials.
- [C.2] **Design of Biomimetic Robotic Eel with Omnidirectional Wire-Driven Body and Controlled Central Pattern Generator.**
Ziyi Sun, et al.
IEEE International Conference on Computer, Control and Robotics (ICCCR), 2025.
Presented the mechanical design and control framework of a biomimetic robotic eel with an omnidirectional quadruple wire-driven body and CPG-based locomotion control.

[S.1] **CHEROE: Every Humanoid Skill as a Traject.**
Ziyi Sun, Jingwen Chen, Yuxi Wang, Xiuze Xia, Long Cheng, Zhaoxiang Zhang, Junyu Dong.
Under review at AAAI 2027 (CCF-A).
Unified SkillMotion representation and Tracker interface with boundary compatibility assessment, trajectory fusion, and bridging-action retrieval for reusable multi-source humanoid skills.

[S.2] **A Streaming Evaluation Framework for Deep-Sea Mining: Comparing Centralized and Distributed Nodule Collection.**
Ziyi Sun, Changhong He.
Under review at *Journal of Marine Science and Engineering (JMSE)*.
Developed a unified simulation and benchmarking pipeline for scalable robot learning, integrating CAD-based robot models and reproducible comparisons of centralized vs. distributed multi-AUV nodule collection.

[P.1] **A High-Stability PTO System for Duck-Type Wave Energy Converters.**
Ziyi Sun, et al.
Invention Patent (First Student Inventor).

[SW.1] **Underwater Polarized Light Field Prediction Software for Biomimetic Navigation V1.0.**
Ziyi Sun.
Software Copyright (No. 2024SR0140235).

SKILLS

- **Robot Platform:**Unitree G1
- **Programming & ML:**Python, C/C++, MATLAB, PyTorch, Isaac Gym/Lab, MuJoCo
- **Mechanical Design:**SolidWorks, AutoCAD
- **Languages:**Chinese (Native); English (CET-4: 591, CET-6: 482)

HONORS AND AWARDS

• Shandong Provincial Government Scholarship	2025
• First-Class Comprehensive Scholarship, Ocean University of China	2024 & 2025
• Outstanding Student / Outstanding Student Cadre, Ocean University of China	2024 & 2025
• National First Prize, National 3D Digital Innovation Design Competition	2025 & 2026
• National Second Prize, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)	2025
• Honorable Mention, Mathematical Contest in Modeling (MCM)	2025
• National Third Prize, 18th National College Student Energy Conservation Competition	2024
• National Gold Award, 3rd “Chuangyi Cup” National Undergraduate Extracurricular Academic Contest	2024
• National Second Prize, 14th Marine Vehicle Design and Production Competition	2025
• National First Prize, CRRC Cup National Undergraduate Renewable Energy Competition	2026
• Science and Technology Innovation Pioneer Team, College of Engineering, OUC	2024

LEADERSHIP EXPERIENCE

• Class Representative / Class Monitor, Ocean University of China	2023 – Present
• Team Leader, College of Engineering Debate Team, OUC	2023 – Present
• Director of Event Organization, Student Association for the Study of Socialism with Chinese Characteristics, OUC	2023 – Present

VOLUNTEER EXPERIENCE

• Volunteer, Qingdao Marathon	2023
• Member, National College Student Volunteer Outreach Team (Zunyi Conference)	2024
• Outstanding Individual in Summer Social Practice, Ocean University of China	2024